Gaël BOYA

Digital Electronic Design Engineer (FPGA/ASIC/SoC)

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France
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LANGUAGES

French
Native language

English
Level C1

SOCIAL NETWORKS

Adébayo BOYA
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SKILLS

Design and Verification
Verilog, SystemVerilog VHDL, RTL design, Icarus Verilog, GTKWave, SVA Assertions UVM Concepts

EDA Tools & Simulation
Vivado (Xilinx), ModelSim, Quartus, CatapultC (HLS)

Architecture & Protocols
RISC-V (RV32I), SoC Integration, Wishbone bus, AXI4-Stream, AXI DMA, UART, SPI, I2C

Programming
C/C++, Python

Versionning
GIT, SVN

SOFT SKILLS

Attention to detail

Analytical mindset

Autonomy & initiative

Team collaboration

Technical documentation

Dual-degree engineer (Grenoble-INP Phelma & INPT) specializing in RTL design and hardware architecture. Industry experience in SoC integration at Silicon Mobility and personal projects covering full RISC-V processor design from scratch. Proficient in Verilog/SystemVerilog digital design, functional verification, and HW/SW co-design on Zynq platforms. Seeking a position in digital design or verification (FPGA/ASIC).

EXPERIENCES

Embedded Systems & IoT Trainer (Freelance)

From March 2026 to April 2026 SUPINFO Lyon, France

- Delivered a 35h intensive course: microcontrollers, sensors, communication protocols (UART, I2C) and digital logic circuits.

SoC Integration Intern Engineer

From February 2024 to August 2024 Silicon Mobility Valbonne, France

- RTL design & top-level integration of a reference SoC (FPCU): architecture decisions, IP block integration, exhaustive RTL simulation and functional verification.
- Improved internal SoC integration tooling and authored technical documentation (user guide), streamlining future project delivery timelines.

Engineering Intern in Architecture & Security

From July 2023 to August 2023 Institut National des Postes et Télécommunications Rabat, Morocco

- Microarchitecture modelling of CPU/SoC designs using gem5 simulator; study, simulation and countermeasure implementation against side-channel attacks (Spectre/Meltdown, Flush+Reload).

EDUCATION

Engineering Degree — Integrated Electronic Systems

From September 2023 to April 2025 Grenoble-INP PHELMA Grenoble, France

Engineering Degree — Embedded Systems and Digital Services

From October 2021 to July 2025 Institut National des Postes et Télécommuniations (INPT) Rabat, Morocco

PROJECTS

RISC-V CPU Design — MiniSoC-RV32I

Since August 2025 Personal Grenoble, France

- Designed from scratch a pipelined 5-stage RV32I core in Verilog: ALU, 32-register file, full hazard (forwarding, stall, flush) and static branch prediction.
- Integrated Wishbone bus, custom peripherals (UART, GPIO, Timer), complete testbench environement (Icarus Verilog + GTKWave) and RISC-V GCC software toolchain.

Hardware Sobel Edge Detector on FPGA

February 2026 Personal Grenoble, France

- AXI4-Stream real-time video pipeline: 3x3 pipelined Sobel filter (3 stages, 125 MHz) with dual line buffers and full backpressure management (*tvalid/tready*).
- Custom SPI controller for SSD1331 OLED display (96x64 RGB565); on-the-fly pattern generation (no external memory); hardware bypass/filter mux via physical switches.

FPGA UART Controller with HIL Verification

February 2026 Personal Grenoble, France

- Verilog UART controller (16x oversampling, 4-stage FSM, CDC reset synchronizer); testbench (SystemVerilog) with concurrent SVA assertions and randomization.
- Hardware-in-the-Loop validation: ESP32 ↔ FPGA Arty Z7 transparent bridge via Pmod (C++ PlatformIO); LED pulse extender for real-time visual feedback.